

Lecture 4b

Wavefront Coding for Extended-Depth-of-Field



Nelson Diaz¹

¹ Department of Computer Science, Universidad Técnica Federico Santa María, Valparaíso, Chile

nelson.diazd@usm.cl

<https://nelson-diaz.com/>

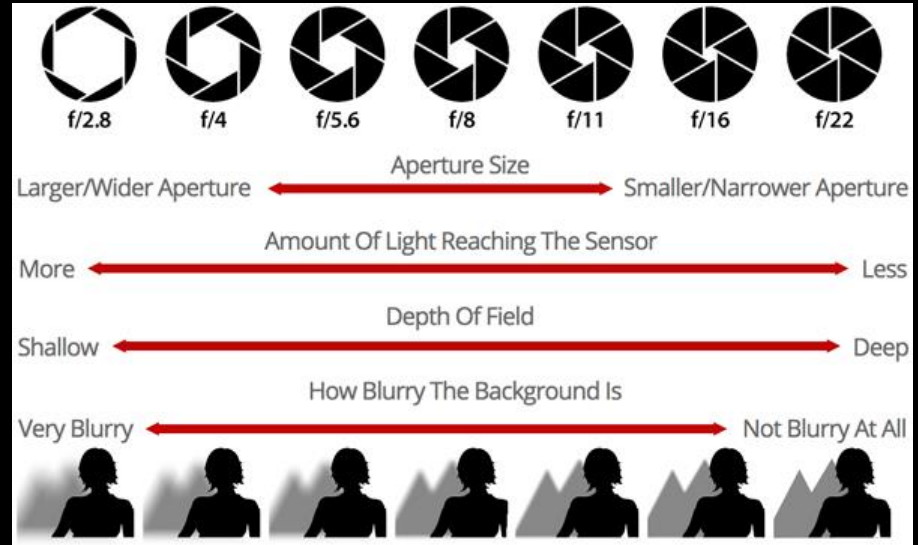


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FEDERICO SANTA MARIA



Depth-of-Field (EDoF)

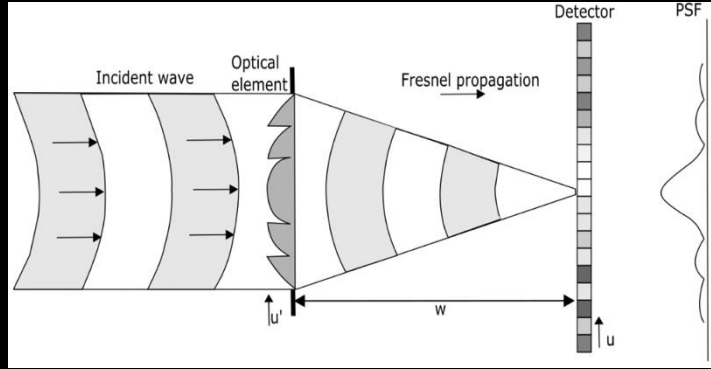
The **depth-of-the-field** refers to the depth range of an object that can be **sharply imaged** by a given optical imaging system and is determined by **operating wavelength**, **effective focal length** and **aperture size** of the imaging lens.



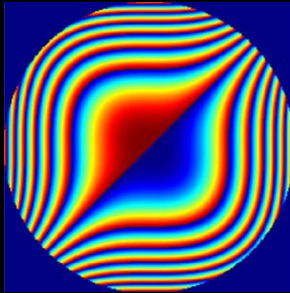
- Reducing the aperture size improves the depth of field
- But reduces the incoming light

Extended-Depth-of-Field (EDoF)

Two assumption when designing EDoF

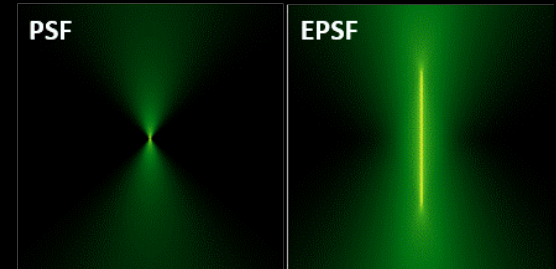


2. It is assumed that any resulting image will be an **intermediate image** and will require **digital processing**¹



1. The **incoherent optical system** is being modified by a rectangular separable **phase mask**¹

Standard vs Extended PSF

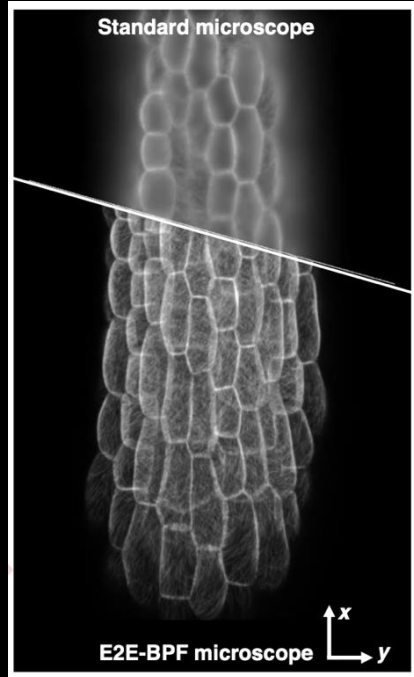


- Conventional cameras has narrow PSF
- The idea of EDoF is to promote PSF insensitive to misfocus

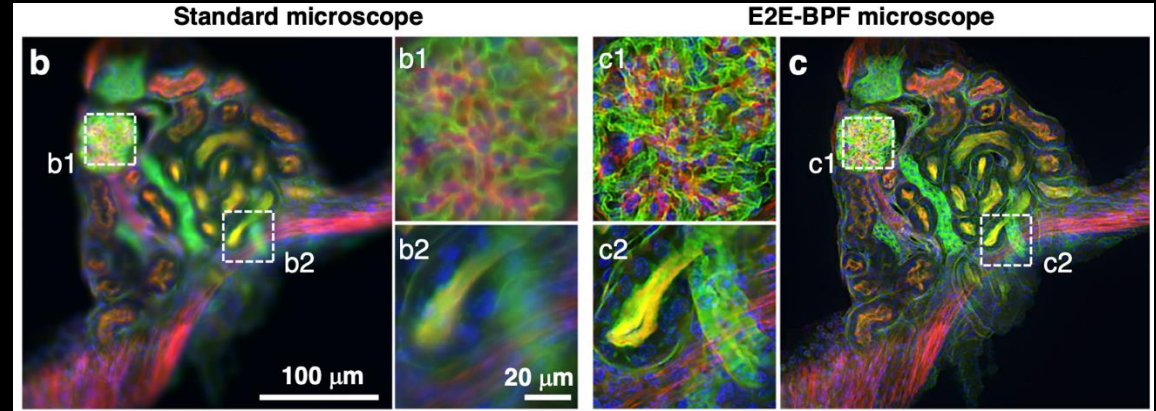
1. E. Dowski and W. Cathey, "Extended depth of field through wave-front coding," Appl. Opt. 34, 1859-1866 (1995).

Applications of EDoF

High-resolution imaging of large-scale objects without serial refocusing is challenging



E2E-BPF microscope¹

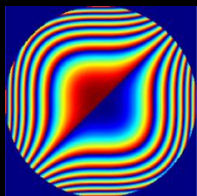


Traditional microscope vs EDoF microscope²

It is crucial to design the phase modulation

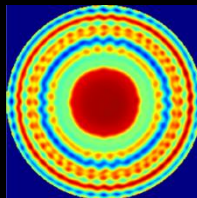
2. Seong, B., Kim, W., Kim, Y. et al. E2E-BPF microscope: extended depth-of-field microscopy using learning-based implementation of binary phase filter and image deconvolution. Light Sci Appl 12, 269 (2023).

Timeline of EDoF through Wavefront Coding



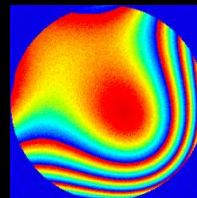
1995

Cubic Phase¹



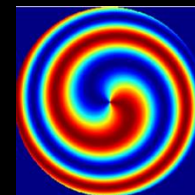
2021

End-to-end optimization³



2023

Hardware-in-the-loop⁴



2024

Our approach⁴

$$h_{\lambda}(s, t) = \text{mod}(\alpha\pi(s^3 + t^3), 2\pi), \\ |s| < 1, |t| < 1$$



f/8



f/1

$$d(s, t) = e^{(jL\theta_{(s,t)} + j\frac{2\pi}{\lambda}(f - \sqrt{f^2 - (s^2 + t^2)}))}$$

Helical Axicon

1. E. Dowski and W. Cathey, "Extended depth of field through wave-front coding," Appl. Opt. 34, 1859-1866 (1995).
3. Akpinar, E. Sahin, M. Meem, R. Menon and A. Gotchev, "Learning Wavefront Coding for Extended Depth of Field Imaging," in IEEE Transactions on Image Processing, vol. 30, pp. 3307-3320, 2021.
4. Pinilla, S., Fröch, J. E., Miri Rostami, S. R., Katkovnik, V., Shevkunov, I., Majumdar, A., & Egiazarian, K. (2023). Miniature color camera via flat hybrid meta-optics. Science Advances, 9(21), eadg7297.

Dowski-Cathey (1995)

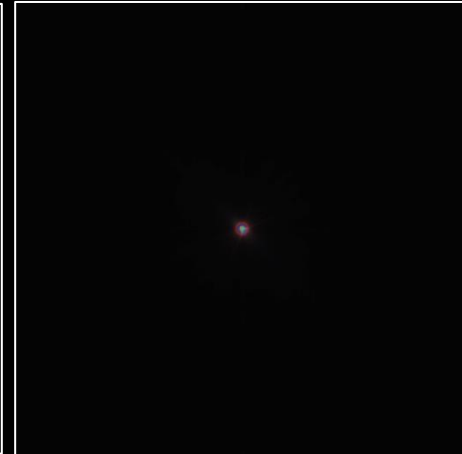
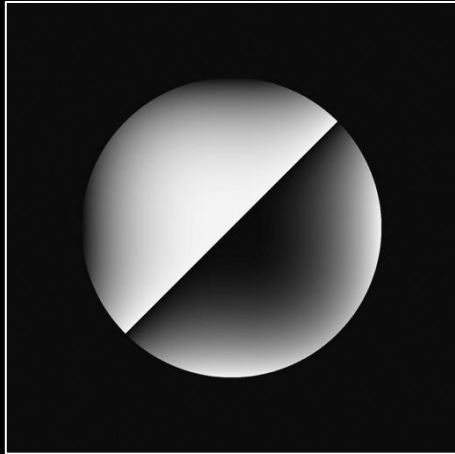
$$h_{\lambda}(s,t) = \text{mod}(\alpha\pi(s^3 + t^3), 2\pi), |s| < 1, |t| < 1$$

$\alpha=[6,16,63,126]$

$z = 0.5\text{m}$

$z = 1.0\text{m}$

$z = 2.0\text{m}$

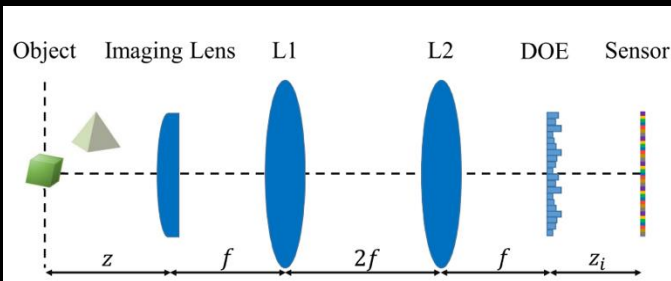


Point spread function at different distances

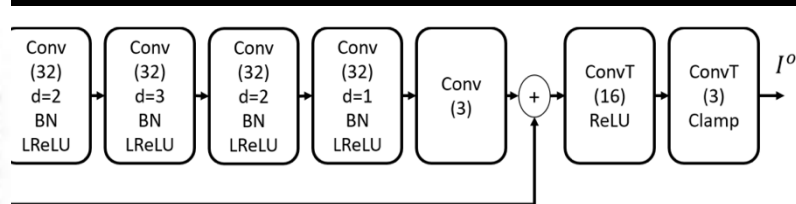
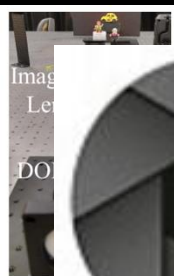
Dowski-Cathey promotes PSF insensitive to misfocus¹

1.E. Dowski and W. Cathey, "Extended depth of field through wave-front coding," Appl. Opt. 34, 1859-1866 (1995).

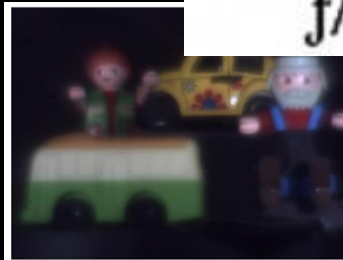
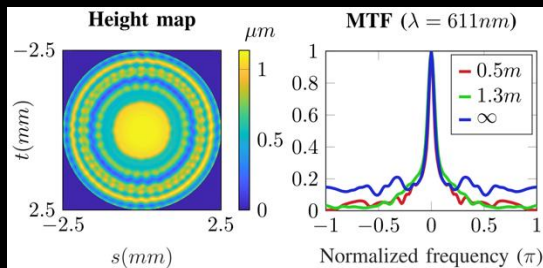
End-to-End Optimization (2021)



Optical setup



CNN-deblurring network



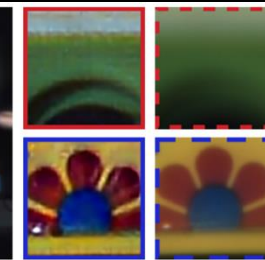
Sensor image



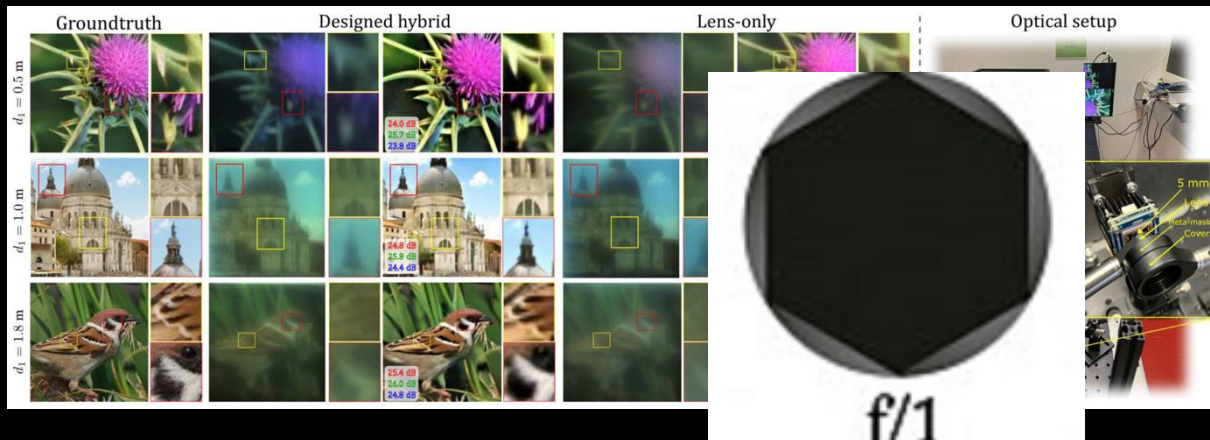
Output of D-CNN³



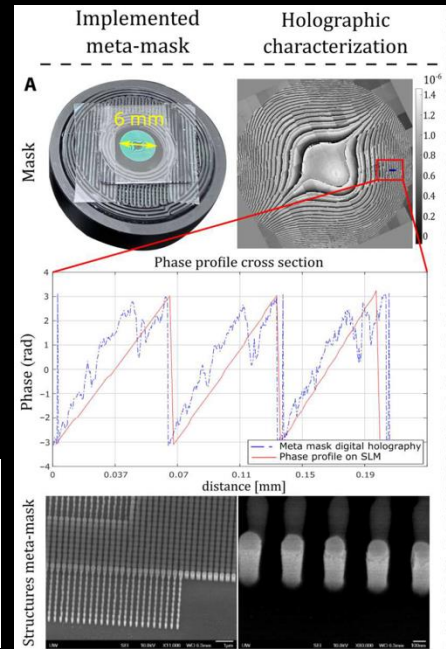
Refractive lens-only



Hardware-in-the-loop^{4,6} (2023)



$$\varphi_{\lambda_0}(x, y) = \frac{2\pi\alpha}{\lambda_0 f \lambda_0} (x^2 + y^2) + \beta(x^3 + y^3) + \sum_{p=1, p \neq 4}^P \rho_p Z_p(x, y),$$



4. Pinilla, S., Frösch, J. E., Miri Rostami, S. R., Katkovnik, V., Shevkunov, I., Majumdar, A., & Egiazarian, K. (2023). Miniature color camera via flat hybrid meta-optics. *Science Advances*, 9(21), eadg7297.

6. Samuel Pinilla, Seyyed Reza Miri Rostami, Igor Shevkunov, Vladimir Katkovnik, and Karen Egiazarian, "Hybrid diffractive optics design via hardware-in-the-loop methodology for achromatic extended-depth-of-field imaging," *Opt. Express* 30, 32633-32649 (2022)

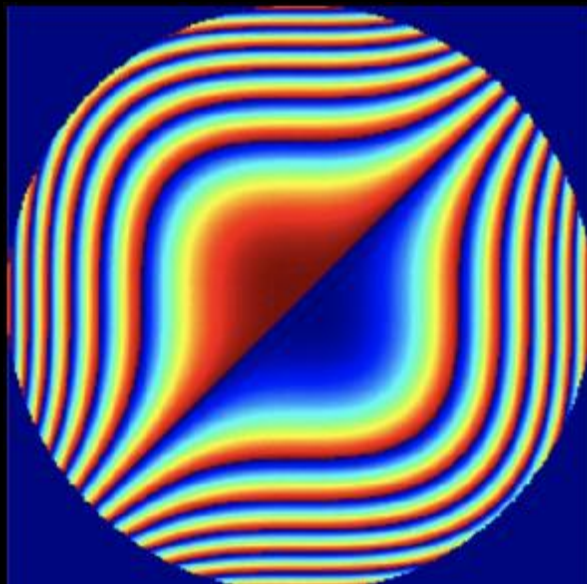
EDoF Discrete Model

$$Y_{RGB,z} = S_{RGB}(\textcolor{red}{PSF}_{\lambda,z} \odot X_{\lambda,z}) + \Omega_{\lambda,z}$$

designing the encoding is critical
to facilitate the inverse problem



measurement



DOE



Scene

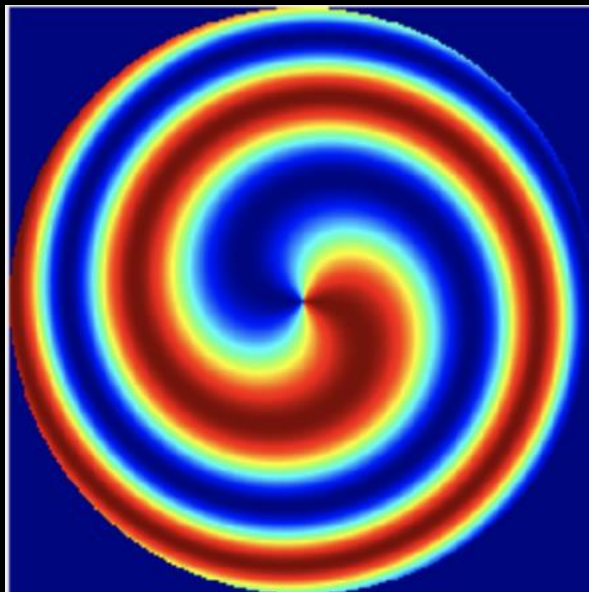
Helical Axicon Wavefront Coding

$$Y_{RGB,z} = S_{RGB}(PSF_{\lambda,z} \odot X_{\lambda,z}) + \Omega_{\lambda,z}$$

novel DOE design



measurements



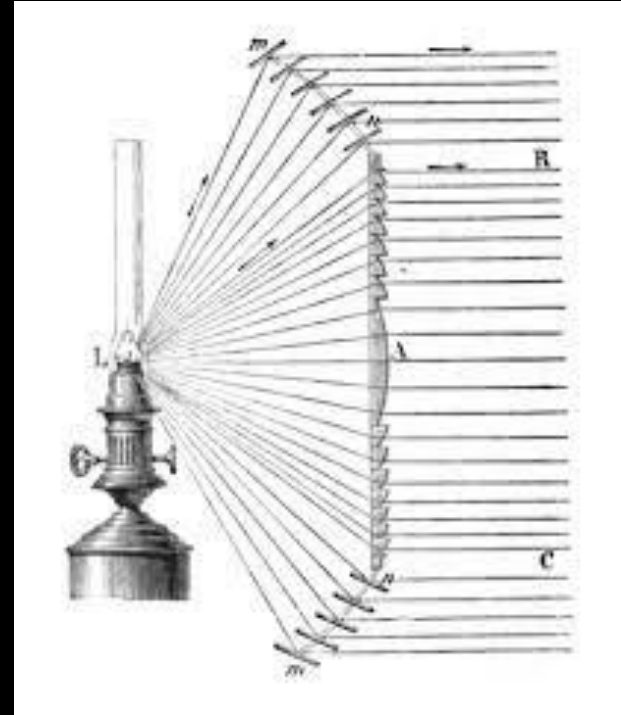
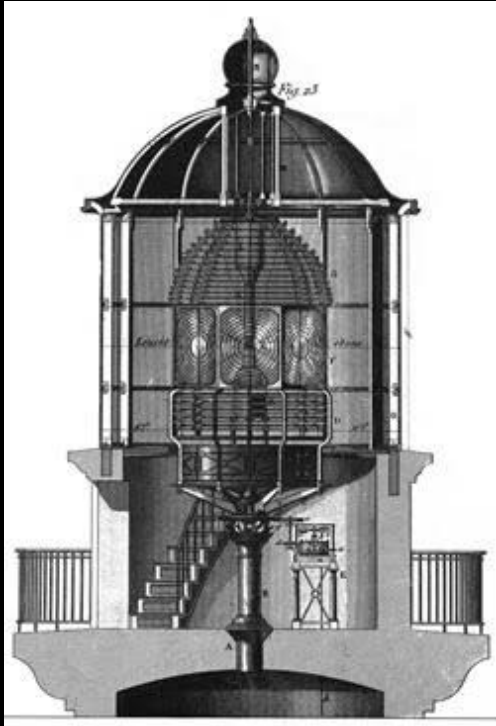
Helical Axicon



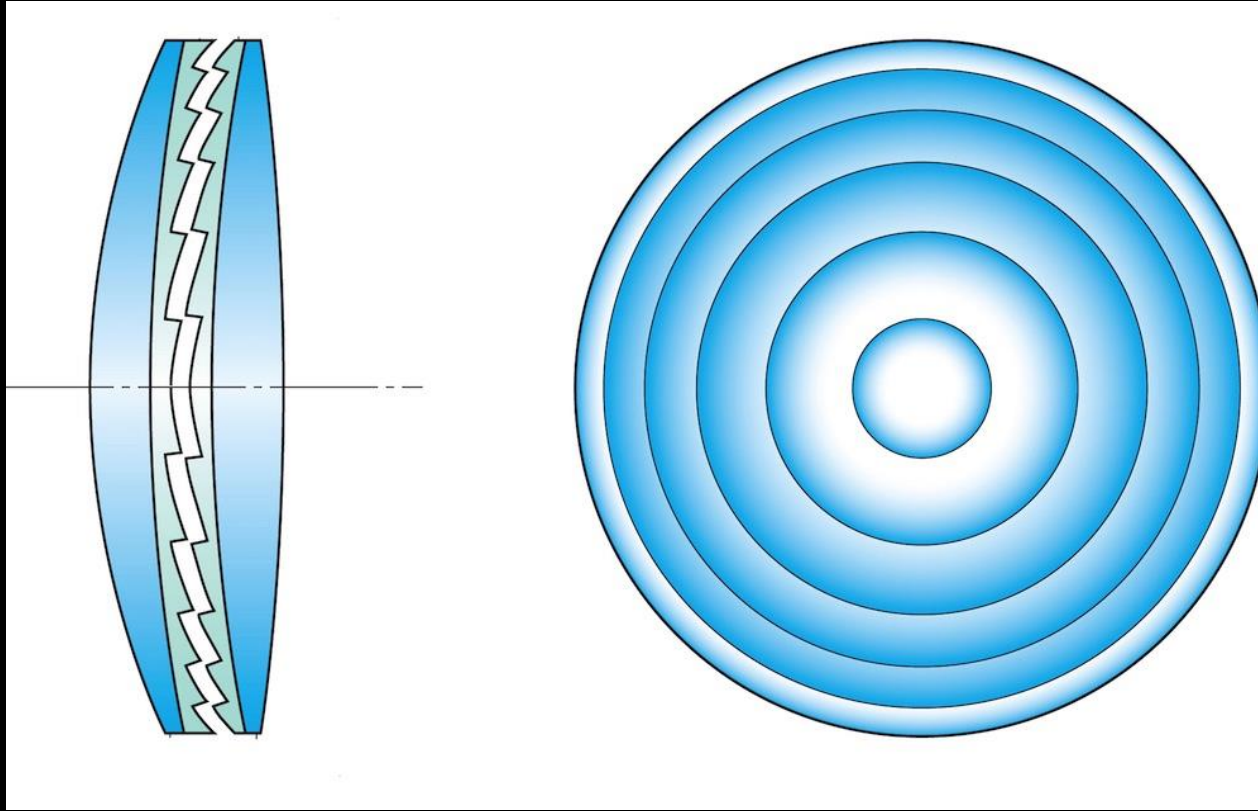
scene

Fresnel Lens

In 1822, French Physicist, **Augustin Fresnel** invented a lens that was able to capture **83% of the light sent to the lens** while flames on reflectors, which were standard at the time, **lost 83% of their light**.



Diffractive Optical Element



7. John H. McLeod, "The Axicon: A New Type of Optical Element," J. Opt. Soc. Am. 44, 592-597 (1954).

Mathematical Model

$$PSF_{\lambda,z} = \left| \mathcal{F}\{Q_{\lambda,z}(s,t)\}_{\frac{x}{\lambda z_i}, \frac{y}{\lambda z_i}} \right|^2,$$

$\mathcal{F}\{.\}$ is the Fourier transform. Where the generalized pupil function is

$$Q_{\lambda,z}(s,t) = A(s,t)e^{j\Phi_{\lambda}(s,t) + j\Psi_{\lambda,z}\left(\frac{s^2+t^2}{r^2}\right)},$$

The thickness function of the diffractive element $\Phi_{\lambda}(s,t) = k(n_{\lambda} - 1)d(s,t),$

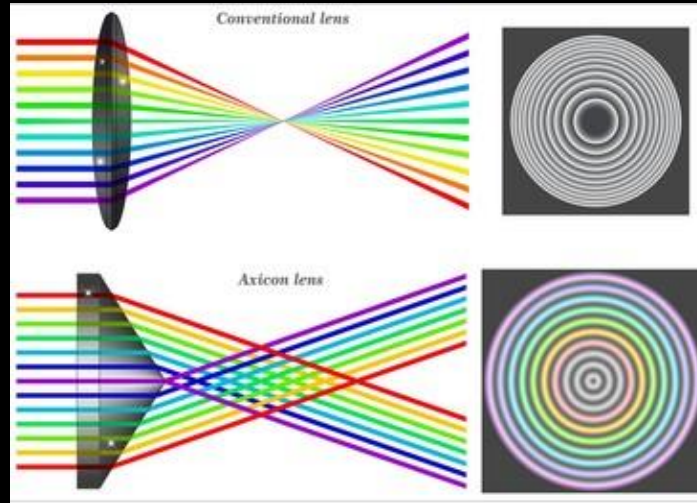
$k=2\frac{\pi}{\lambda}$ is the wavenumber and n_{λ} is the wavelength-dependent refractive index.

Where the defocus coefficient is $\Psi_{\lambda,z} = \frac{\pi}{\lambda} \left(\frac{1}{z} + \frac{1}{z_i} - \frac{1}{f_{\lambda}} \right) r^2$

f_{λ} is the wavelength-dependent focal, r is the radius of the pupil, z distance of the object and z_i is the distance from the DOE to the RGB sensor.

Axicon

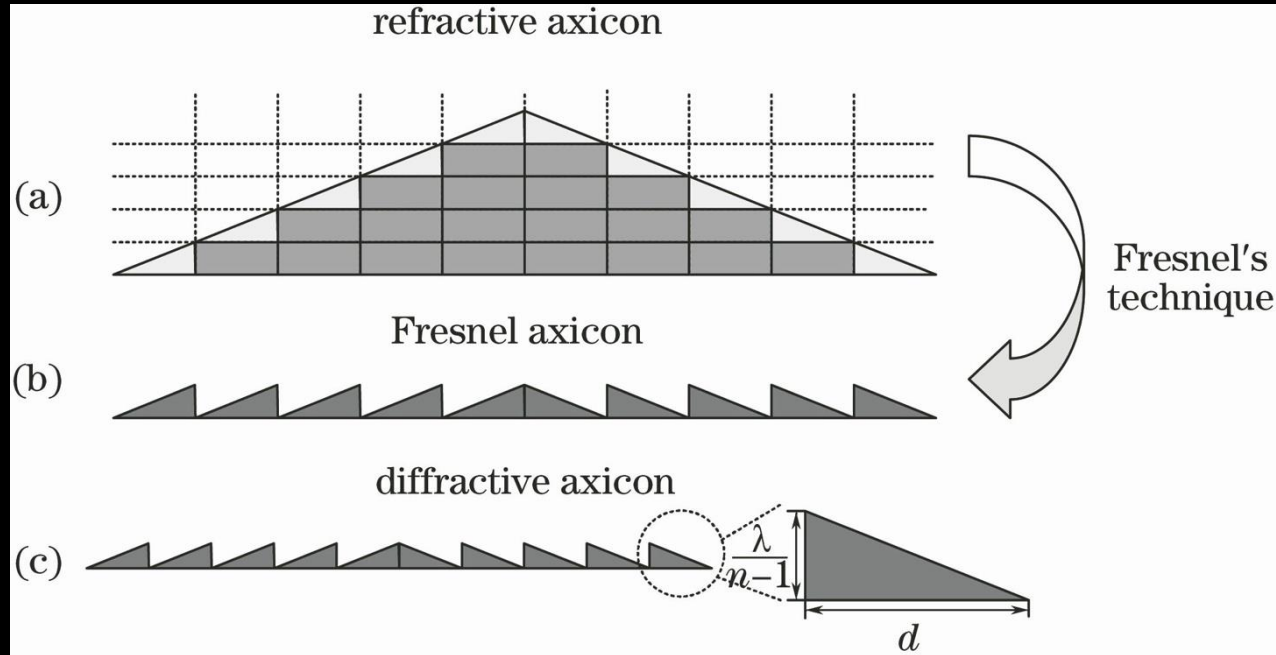
- An axicon is a specialized type of lens that has a conical surface. An axicon transforms a laser beam into a **ring shaped** distribution.



One application of axicons is in **telescopes**, where the spherical objective is replaced by an axicon⁷. Such a telescope can be **simultaneously in focus for targets at distances from less than a meter to infinity**, without making any adjustments.

7. John H. McLeod, "The Axicon: A New Type of Optical Element," J. Opt. Soc. Am. 44, 592-597 (1954).

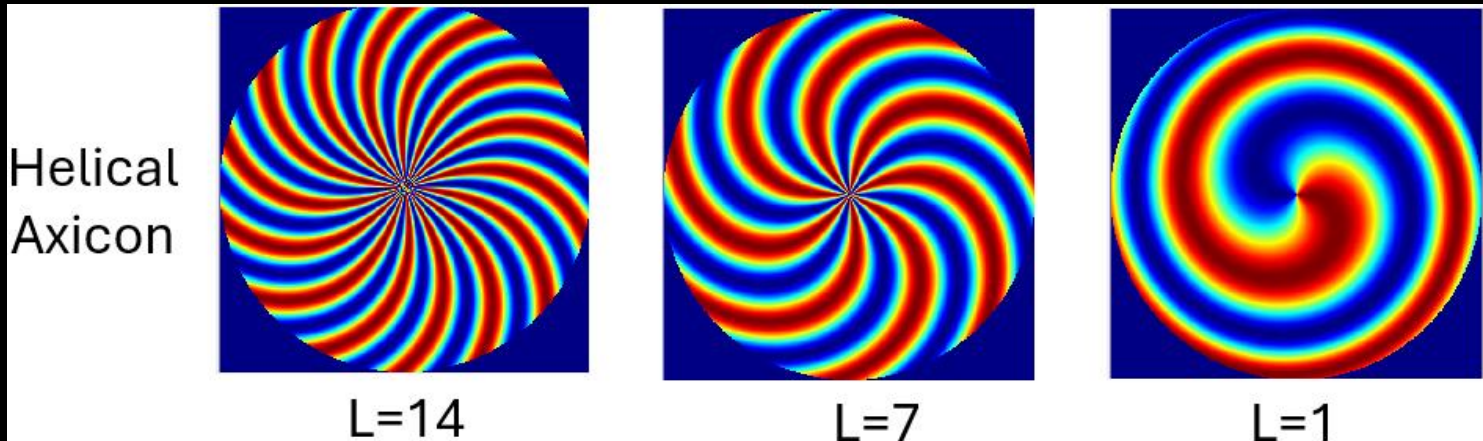
From Axicon to Diffractive Optical Element



Helical Axicon

$$d(s, t) = e^{jL\theta_{(s,t)} + j\frac{2\pi}{\lambda}(f - \sqrt{f^2 - (s^2 + t^2)})}$$

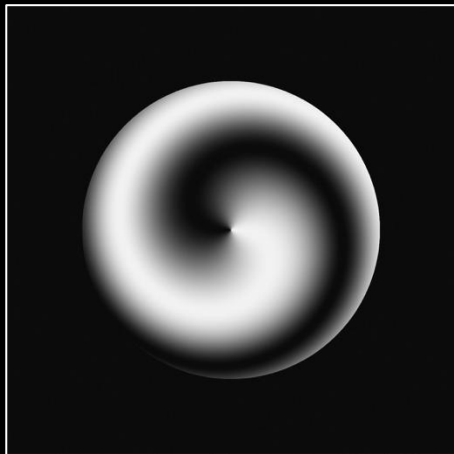
- L is the number of wings
- f is the focal distance
- λ is the wavelength
- $\theta_{(s,t)} = \arctan(s, t)$ the angle between s and t .



Helical Axicon

$$d(s, t) = e^{(jL\theta_{(s,t)} + j\frac{2\pi}{\lambda} (f - \sqrt{f^2 - (s^2 + t^2)}))}$$

$f = [6, 16, 63, 126]$



$z=0.5\text{m}$



$z=1.0\text{m}$



$z=2.0\text{m}$



Our approach improves PSF insensitive to misfocus

Simulation

- 18 RGB images
- Pixels number 500 x 500 x 3
- McMaster dataset⁹
- Reconstruction algorithm: TwIST L2-TV



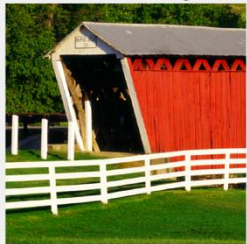
Fabrication Parameters

Parameter	Our implementation
Minimum feature width	1.86 μm
Maximum feature depth	2.4 μm
Number of grayscale	128
Sensor pitch	3.72 μm

9. Lei Zhang, Xiaolin Wu, Antoni Buades, and Xin Li "Color demosaicking by local directional interpolation and nonlocal adaptive thresholding," Journal of Electronic Imaging 20(2), 023016 (1 April 2011). <https://doi.org/10.1117/1.3600632>.

Dowski-Cathey (1995)

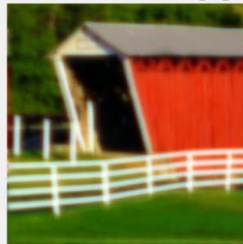
Groundtruth Image 18



Measurement $z=0.5[m]$



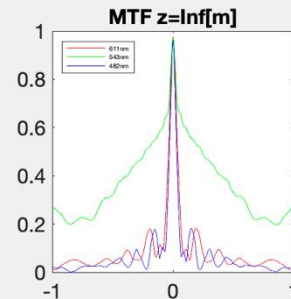
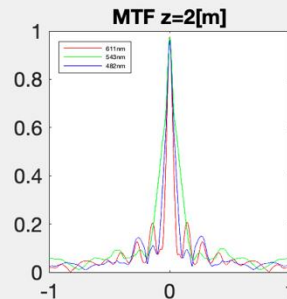
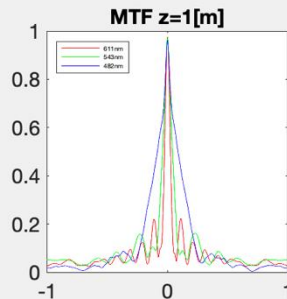
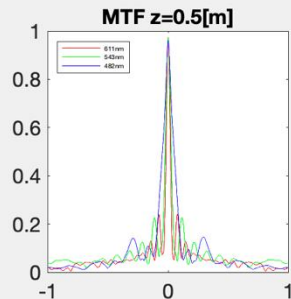
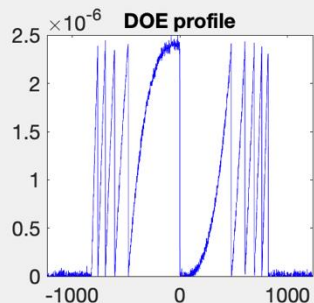
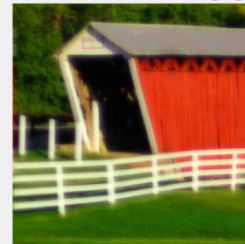
Measurement $z=1[m]$



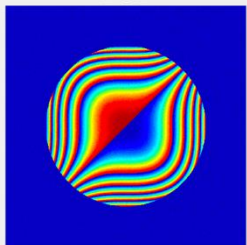
Measurement $z=2[m]$



Measurement $z=\text{Inf}[m]$



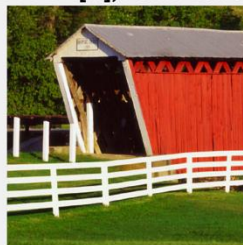
DOE



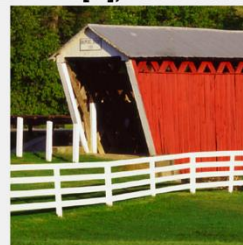
I2-TV $z=0.5[m]$, PSNR=27.9519



I2-TV $z=1[m]$, PSNR=29.7855



I2-TV $z=2[m]$, PSNR=30.0651



I2-TV $z=\text{Inf}[m]$, PSNR=32.988

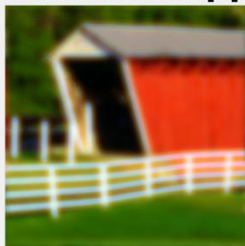


End-to-End (2021)

Groundtruth Image 18



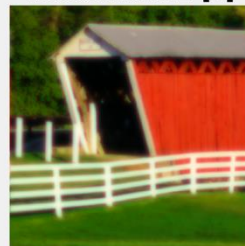
Measurement $z=0.5$ [m]



Measurement $z=1$ [m]



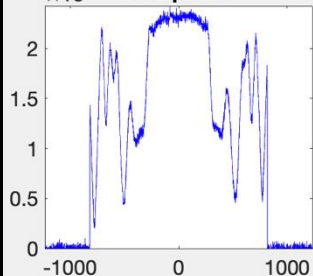
Measurement $z=2$ [m]



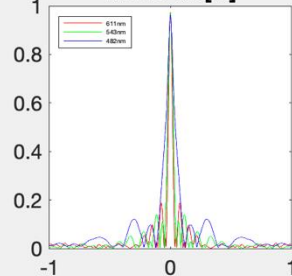
Measurement $z=\text{Inf}$ [m]



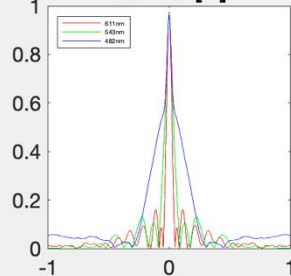
$\times 10^{-6}$ DOE profile



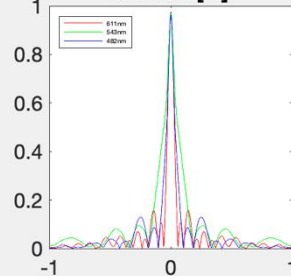
MTF $z=0.5$ [m]



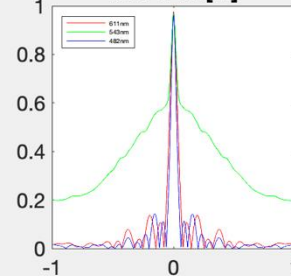
MTF $z=1$ [m]



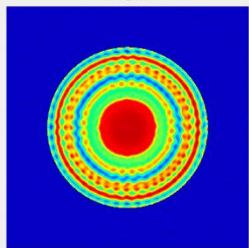
MTF $z=2$ [m]



MTF $z=\text{Inf}$ [m]



DOE



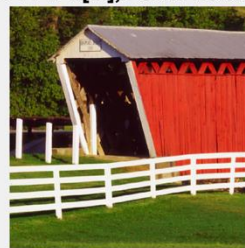
I2-TV $z=0.5$ [m], PSNR=25.3168



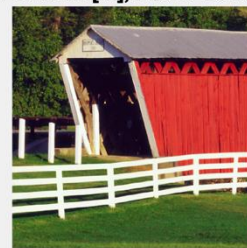
I2-TV $z=1$ [m], PSNR=26.1885



I2-TV $z=2$ [m], PSNR=28.0293

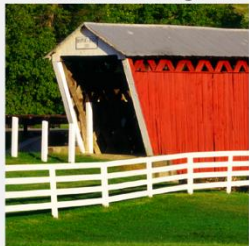


I2-TV $z=\text{Inf}$ [m], PSNR=26.7352



Hardware-in-the-loop (2023)

Groundtruth Image 18



Measurement $z=0.5[m]$



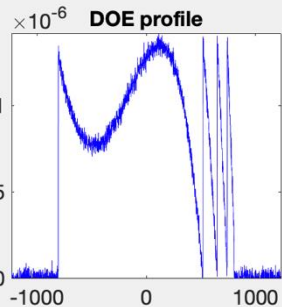
Measurement $z=1[m]$



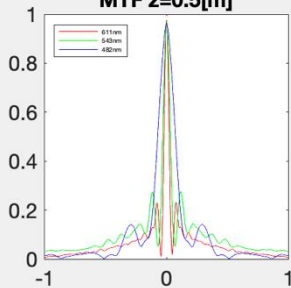
Measurement $z=2[m]$



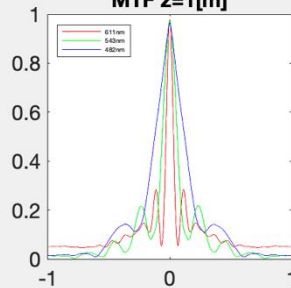
Measurement $z=Inf[m]$



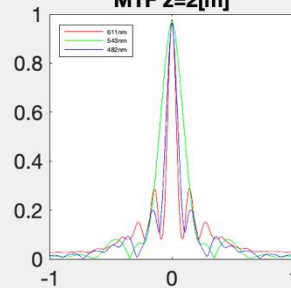
MTF $z=0.5[m]$



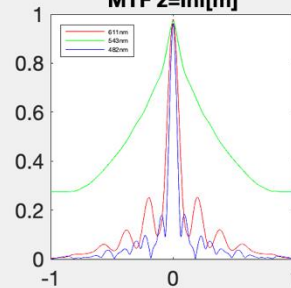
MTF $z=1[m]$



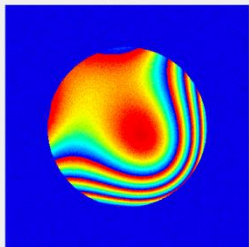
MTF $z=2[m]$



MTF $z=Inf[m]$



DOE



I2-TV $z=0.5[m]$, PSNR=30.7055



I2-TV $z=1[m]$, PSNR=33.6891



I2-TV $z=2[m]$, PSNR=33.4791

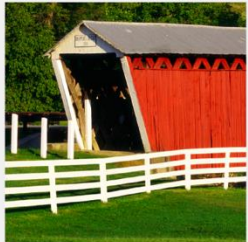


I2-TV $z=Inf[m]$, PSNR=32.1636



Our Helical Axicon

Groundtruth Image 18



Measurement $z=0.5[m]$



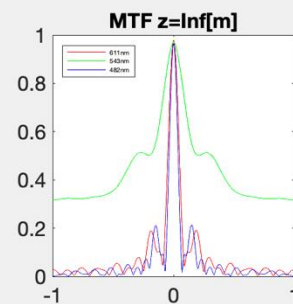
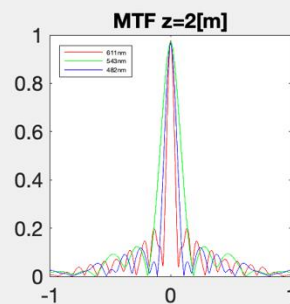
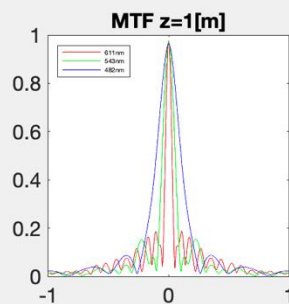
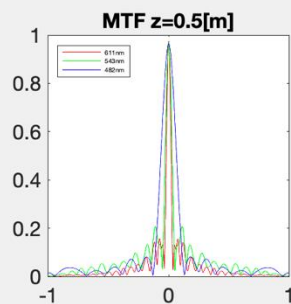
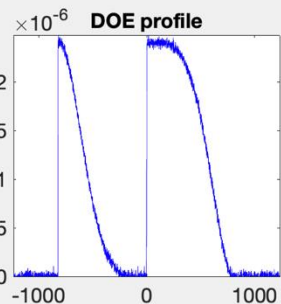
Measurement $z=1[m]$



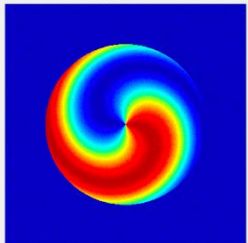
Measurement $z=2[m]$



Measurement $z=Inf[m]$



DOE



I2-TV $z=0.5[m]$, PSNR=31.6823



I2-TV $z=1[m]$, PSNR=30.8641



I2-TV $z=2[m]$, PSNR=34.0705



I2-TV $z=Inf[m]$, PSNR=33.5377



Simulation results

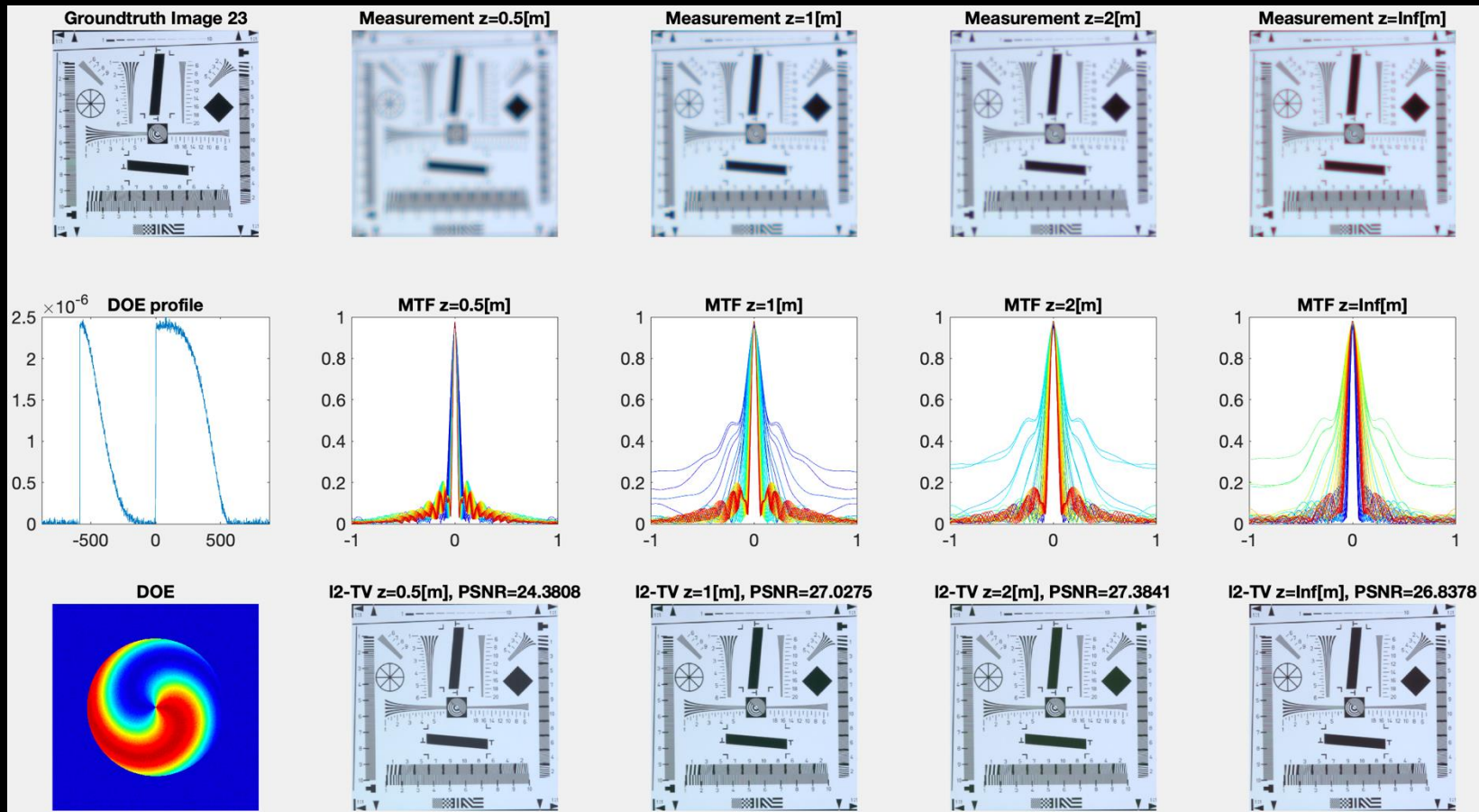
- 18 RGB images
- Pixels number 500 x 500 x 3
- McMaster dataset⁹
- Reconstruction algorithm: TwIST L2-TV



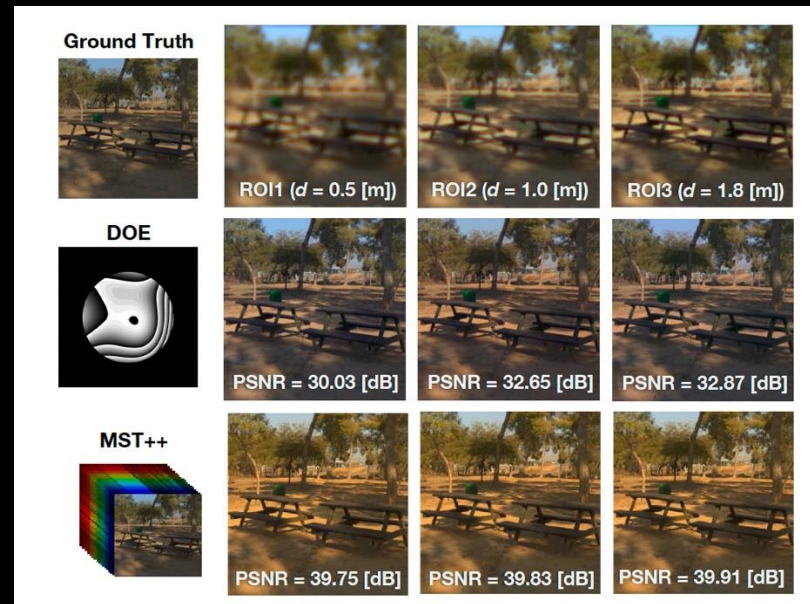
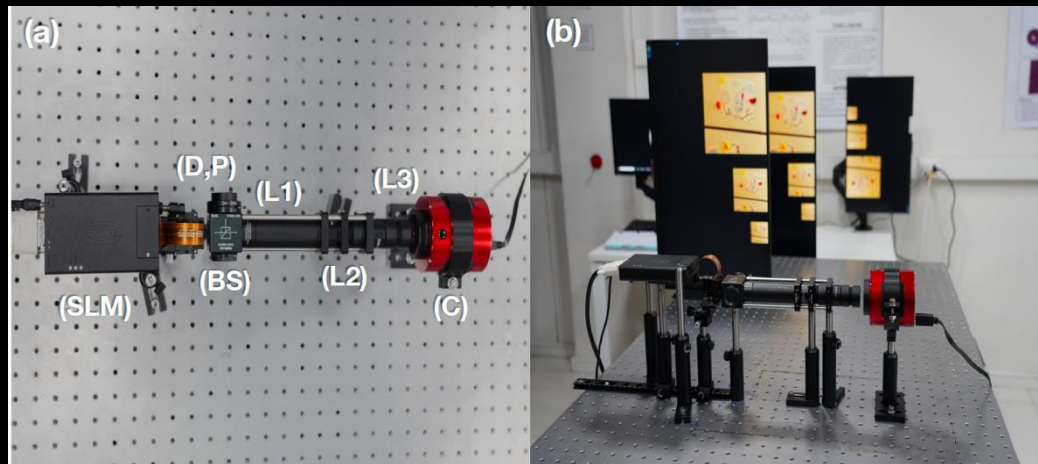
TwIST L2-TV	$\sigma = 0.005$		$\sigma = 0.020$	
DoE \ Metric	PSNR	SSIM	PSNR	SSIM
E2E TIP 2021	24.9501 ± 4.4193	0.7775 ± 0.1985	25.0395 ± 4.5437	0.7753 ± 0.1989
Dowski-Cathey	26.0917 ± 5.1242	0.8067 ± 0.1993	26.1358 ± 5.1561	0.8082 ± 0.2019
HIL, 2023	27.1157 ± 5.2514	0.8341 ± 0.1993	27.2286 ± 5.3389	0.8370 ± 0.1951
Helical Axicon	27.3152 ± 5.4874	0.8384 ± 0.189	27.2645 ± 5.4565	0.8384 ± 0.1891

9. Lei Zhang, Xiaolin Wu, Antoni Buades, and Xin Li "Color demosaicking by local directional interpolation and nonlocal adaptive thresholding," Journal of Electronic Imaging 20(2), 023016 (1 April 2011). <https://doi.org/10.1117/1.3600632>.

Our Helical Axicon with Multispectral Image



Experimental Implementation



Please visit Exequiel and Bejamin work's at poster session on
Wednesday, November 27th, 3:30 pm to 5:00 pm

- Pinilla, S., Fröch, J. E., Miri Rostami, S. R., Katkovnik, V., Shevkunov, I., Majumdar, A., & Egiazarian, K. (2023). Miniature color camera via flat hybrid meta-optics. *Science Advances*, 9(21), eadg7297.
- Samuel Pinilla, Seyyed Reza Miri Rostami, Igor Shevkunov, Vladimir Katkovnik, and Karen Egiazarian, "Hybrid diffractive optics design via hardware-in-the-loop methodology for achromatic extended-depth-of-field imaging," *Opt. Express* 30, 32633-32649 (2022)

Conclusions

- The state of the art of EDoF was presented and compared with the proposed **Helical Axicon** design
- The designed **wavefront coding** modulates the scene promoting **PSF insensitive to misfocus**
- The proposed **Helical Axicon** approach outperforms state-of-the-art **wavefront coding**

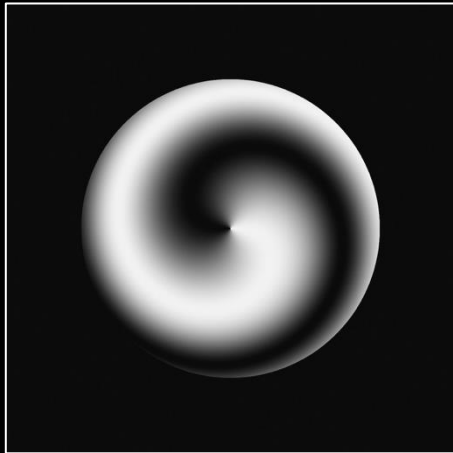
$$d(s, t) = e^{(jL\theta_{(s,t)} + j\frac{2\pi}{\lambda}(f - \sqrt{f^2 - (s^2 + t^2)}))}$$

$f = [6, 16, 63, 126]$

$z = 0.5\text{m}$

$z = 1.0\text{m}$

$z = 2.0\text{m}$



Helical Axicon

Point spread function at different distances

Thanks

ANID ANILLOS (ATE220022); ANID FONDECYT (1221883); ANID FONDECYT Postdoctorado (3230489);
DOCTORADO NACIONAL 2022-21221399.



UNIVERSIDAD TECNICA
FEDERICO SANTA MARIA



nelson.diazd@usm.cl

<https://nelson-diaz.com/>